

מעבדה בתכנון ספרתי מתקדם

3

מגשים:

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צוות 7

התוכנית:

```
7  library IEEE;
8  | use IEEE.STD_LOGIC_1164.ALL;
9  | use IEEE.STD_LOGIC_UNSIGNED.ALL;
10
11  entity REAL_CLK is
12  | port(
13  |   CLK : in std_logic;
14  |   RESET : in std_logic;
15  |   TestMode : in std_logic;
16  |   Seconds_Out: out std_logic_VECTOR[5 downto 0];
17  |   Minutes_Out: out std_logic_VECTOR[5 downto 0];
18  |   Hours_Out: out std_logic_VECTOR[4 downto 0]
19  | );
20
21  end REAL_CLK;
22
23
24  architecture ARC_REAL_CLK of REAL_CLK is
25  | signal Q_Seconds : std_logic_VECTOR(5 downto 0);
26  | signal Q_Minutes : std_logic_VECTOR(5 downto 0);
27  | signal Q_Hours : std_logic_VECTOR(4 downto 0);
28  | signal Q : std_logic_VECTOR(26 downto 0);
29  | signal TC_TimeBase : std_logic;
30  | begin
31  |
32  | -----tc time base and mux for test mode
33  | process
34  | begin
35  |   wait until rising_edge(CLK);
36  |   if (RESET = '1') then
37  |     Q <= (others => '0');
38  |     TC_TimeBase <= '0';
39  |   else if (TestMode = '1') then
40  |     if (Q < 9) then
41  |       Q <= Q + 1;
42  |       TC_TimeBase <= '0';
43  |     else
44  |       Q <= (others => '0');
45  |       TC_TimeBase <= '1';
46  |     end if;
47  |   else if (TestMode = '0') then
48  |     if (Q < 99999999) then
49  |       Q <= Q + 1;
50  |       TC_TimeBase <= '0';
51  |     else
52  |       Q <= (others => '0');
53  |       TC_TimeBase <= '1';
54  |     end if;
55  |   end if;
56  | end process;
57
```

```

61 ----- time counting up to seconds
62 process
63 begin
64     wait until rising_edge(CLK);
65     if (RESET = '1') then
66         Q_Seconds <= (others => '0');
67     else
68         if (TC_TimeBase = '1') then
69             if (Q_Seconds < 59) then
70                 Q_Seconds <= Q_Seconds + 1;
71             else
72                 Q_Seconds <= (others => '0');
73             end if;
74         end if;
75     end if;
76 end process;
77 Seconds_Out <= Q_Seconds;
78
79
80
81
82 ----- time counting up to minutes
83 process
84 begin
85     wait until rising_edge(CLK);
86     if (RESET = '1') then
87         Q_Minutes <= (others => '0');
88     else
89         if ((TC_TimeBase = '1') and (Q_Seconds = 59)) then
90             if (Q_Minutes < 59) then
91                 Q_Minutes <= Q_Minutes + 1;
92             else
93                 Q_Minutes <= (others => '0');
94             end if;
95         end if;
96     end if;
97 end process;
98 Minutes_Out <= Q_Minutes;
99
100
101 ----- time counting up to hours
102 process
103 begin
104     wait until rising_edge(CLK);
105     if (RESET = '1') then
106         Q_HOURS <= (others => '0');
107     else
108         if ((TC_TIMEBASE = '1') and (Q_MINUTES = 59) and (Q_SECONDS = 59)) then
109             if (Q_HOURS < 23) then
110                 Q_HOURS <= Q_HOURS + 1;
111             else
112                 Q_HOURS <= (others => '0');
113             end if;
114         end if;
115     end if;
116 end process;
117 Hours_Out <= Q_Hours;
118
119 end ARC_REAL_CLK;
120
121
122
123
124
125

```

Test bench:

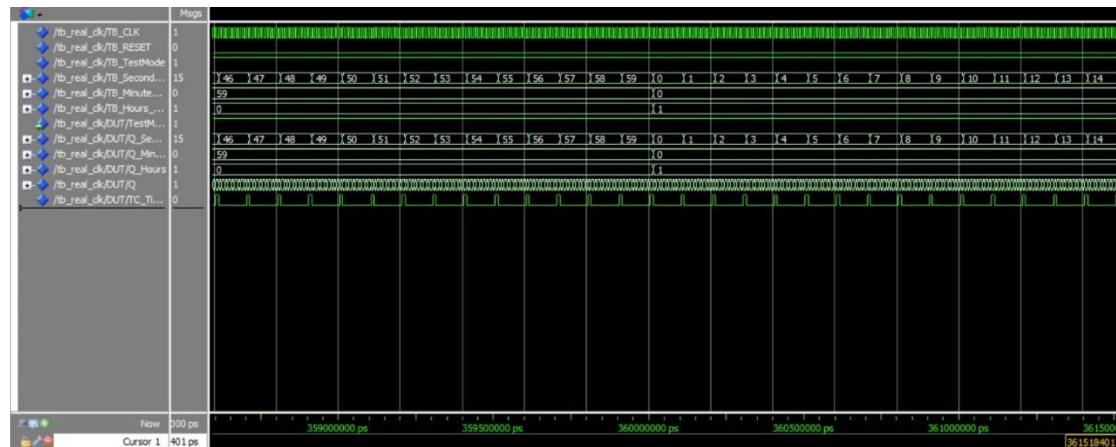
```
11 entity TB_REAL_CLK is
12
13 end TB_REAL_CLK;
14
15 architecture TB_ART_CLK of TB_REAL_CLK is
16 component REAL_CLK
17 port(
18     CLK          : in std_logic;
19     RESET        : in std_logic;
20     TestMode     : in std_logic;
21     Seconds_Out  : out std_logic_VECTOR(5 downto 0);
22     Minutes_Out  : out std_logic_VECTOR(5 downto 0);
23     Hours_Out   : out std_logic_VECTOR(4 downto 0)
24 );
25 end component;
26
27 signal TB_CLK          : std_logic = '0' ;
28 signal TB_RESET        : std_logic;
29 signal TB_TestMode     : std_logic;
30 signal TB_Seconds_Out  : std_logic_VECTOR(5 downto 0);
31 signal TB_Minutes_Out  : std_logic_VECTOR(5 downto 0);
32 signal TB_Hours_Out   : std_logic_VECTOR(4 downto 0);
33
34
35 begin
36
37     DUT : REAL_CLK
38
39 PORT map(
40     CLK          => TB_CLK,
41     RESET        => TB_RESET,
42     TestMode     => TB_TestMode,
43     Seconds_Out  => TB_Seconds_Out,
44     Minutes_Out  => TB_Minutes_Out,
45     Hours_Out   => TB_Hours_Out
46 );
47
48
49 TB_RESET <= '1',
50           '0' after 13 ns;
51 TB_TestMode <= '1';
52
53
54
55 process
56 begin
57     wait for 5 ns;
58     TB_CLK <= not(TB_CLK);
59 end process;
60 -- בשביל שהקלט יפלט כל שניה
61
62
63 end TB_ART_CLK;
```

XDC:

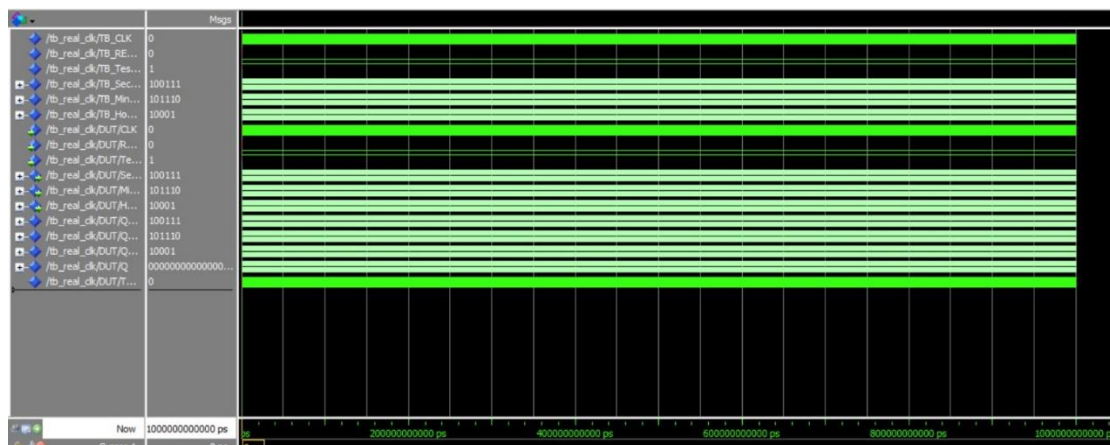
```
1 #####
2 # Mentor Graphics Corporation
3 #
4 #####
5
6
7 #####
8 # create clock #
9 #####
10
11 # Precision Generated
12 create_clock [get_ports {CLK}] -name {CLK} -period 10 -waveform {0 5}
13 create_clock -name {VirtualClock} -period 10 -waveform {0 5}
14
15 #####
16 # set input delay #
17 #####
18
19 # Precision Generated
20 set_input_delay 0 -clock CLK -add_delay [get_ports RESET]
21
22 #####
23 # set output delay #
24 #####
25
26 # Precision Generated
27 set_output_delay 0 -clock CLK -add_delay [get_ports {Seconds_Out* Minutes_Out* Hours_Out*}]
28
29
30 #####
31 # Pin Locations and Voltages #
32 #####
33 set_property -dict { PACKAGE_PIN W5 IOSTANDARD LVCMOS33 } [get_ports { CLK }];
34 set_property -dict { PACKAGE_PIN U18 IOSTANDARD LVCMOS33 } [get_ports { RESET }];
35 set_property -dict { PACKAGE_PIN T17 IOSTANDARD LVCMOS33 } [get_ports { TestMode }];
36
37 set_property -dict { PACKAGE_PIN J1 IOSTANDARD LVCMOS33 } [get_ports { Seconds_Out[0] }];
38 set_property -dict { PACKAGE_PIN L2 IOSTANDARD LVCMOS33 } [get_ports { Seconds_Out[1] }];
39 set_property -dict { PACKAGE_PIN J2 IOSTANDARD LVCMOS33 } [get_ports { Seconds_Out[2] }];
40 set_property -dict { PACKAGE_PIN G2 IOSTANDARD LVCMOS33 } [get_ports { Seconds_Out[3] }];
41 set_property -dict { PACKAGE_PIN H1 IOSTANDARD LVCMOS33 } [get_ports { Seconds_Out[4] }];
42 set_property -dict { PACKAGE_PIN K2 IOSTANDARD LVCMOS33 } [get_ports { Seconds_Out[5] }];
43
44 set_property -dict { PACKAGE_PIN A14 IOSTANDARD LVCMOS33 } [get_ports { Minutes_Out[0] }];
45 set_property -dict { PACKAGE_PIN A16 IOSTANDARD LVCMOS33 } [get_ports { Minutes_Out[1] }];
46 set_property -dict { PACKAGE_PIN B15 IOSTANDARD LVCMOS33 } [get_ports { Minutes_Out[2] }];
47 set_property -dict { PACKAGE_PIN B16 IOSTANDARD LVCMOS33 } [get_ports { Minutes_Out[3] }];
48 set_property -dict { PACKAGE_PIN A15 IOSTANDARD LVCMOS33 } [get_ports { Minutes_Out[4] }];
49 set_property -dict { PACKAGE_PIN A17 IOSTANDARD LVCMOS33 } [get_ports { Minutes_Out[5] }];
50
51 set_property -dict { PACKAGE_PIN K17 IOSTANDARD LVCMOS33 } [get_ports { Hours_Out[0] }];
52 set_property -dict { PACKAGE_PIN M18 IOSTANDARD LVCMOS33 } [get_ports { Hours_Out[1] }];
53 set_property -dict { PACKAGE_PIN N17 IOSTANDARD LVCMOS33 } [get_ports { Hours_Out[2] }];
54 set_property -dict { PACKAGE_PIN P18 IOSTANDARD LVCMOS33 } [get_ports { Hours_Out[3] }];
55 set_property -dict { PACKAGE_PIN L17 IOSTANDARD LVCMOS33 } [get_ports { Hours_Out[4] }];
56
57
58 #####
59 # Configuration Settings #
60 #####
61 set_property CFGBVS VCC0 [current_design]
62 set_property CONFIG_VOLTAGE 3.3 [current_design]
63
64
65
```

Waveforms for

Ex 1:

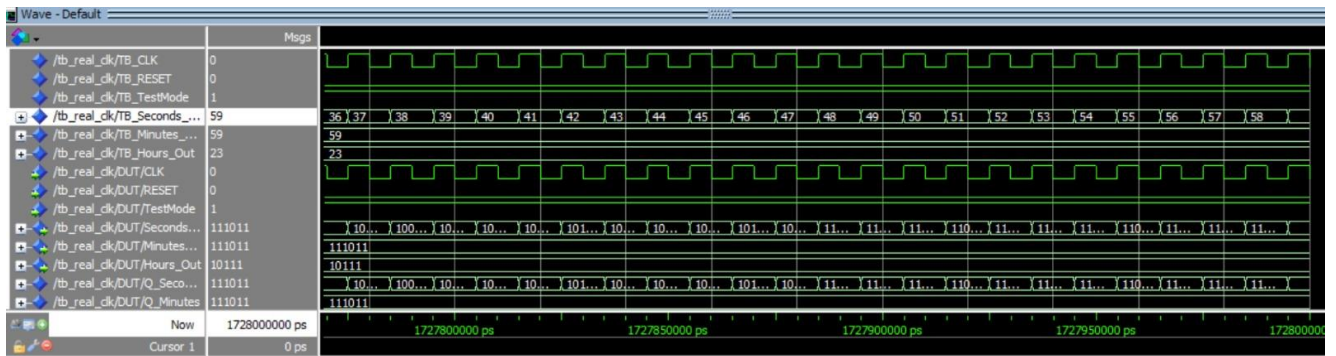


Ex 2:



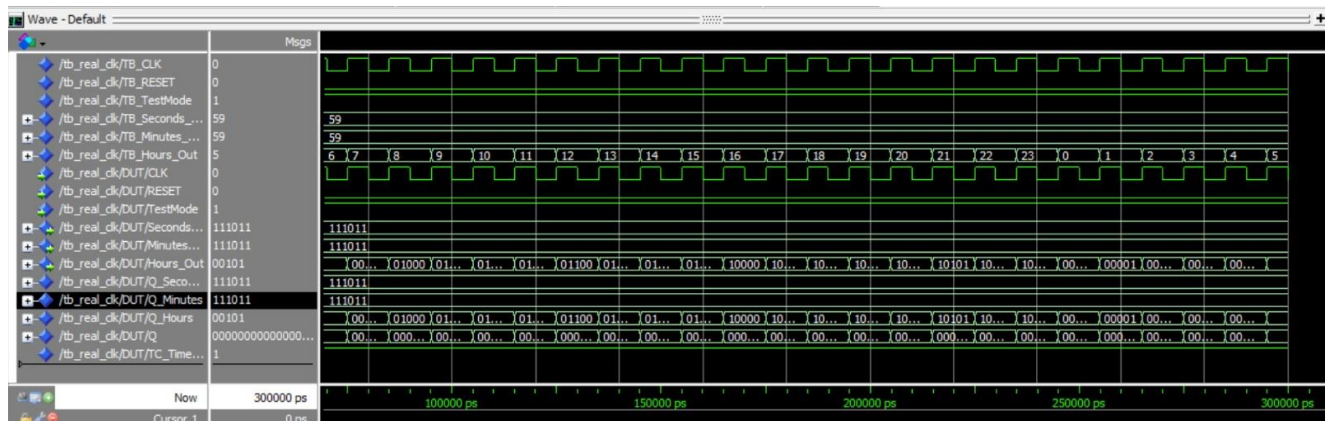
לקח לנו 5:35 דקות אמיתיות כדי לסמלץ שניה אחת
ז"א שאם נסמלץ 24 שעות יקח לנו 480 קילו דקות.

Ex 3:



אפשר לראות שהשניות משתנות כל מחזור שעון.

Ex 4:



אפשר לראות שהשעות משתנות כל מחזור שעון.

4. Timing and area reports

Utilization

Post-Synthesis | Post-Implementation

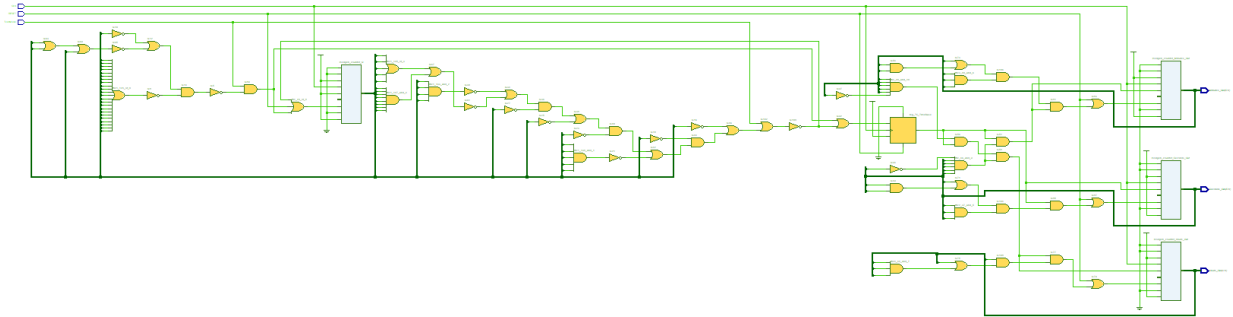
Graph | Table

Resource	Utilization	Available	Utilization %
LUT	32	20800	0.15
FF	45	41600	0.11
IO	20	106	18.87
BUFG	1	32	3.13

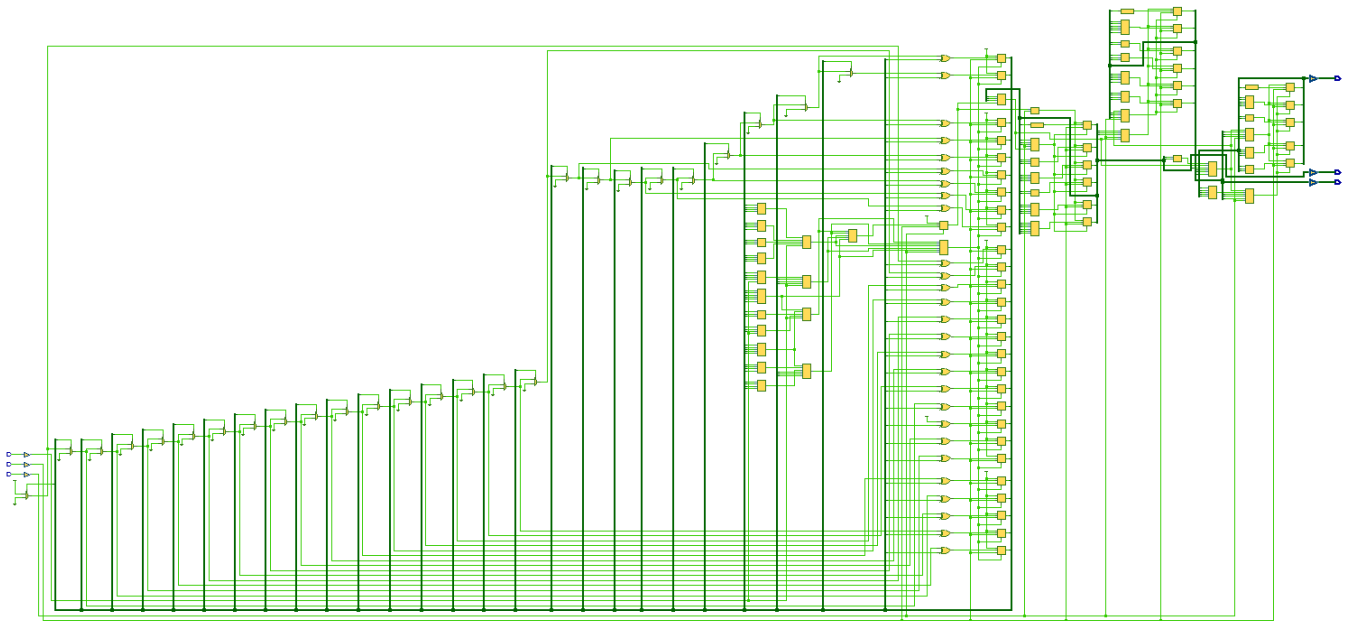
Intra-Clock Paths - CLK - Setup														
Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Uncertainty
↳ Path 1	-0.872	1	1	1	reg_q(0)_1_12_replica/C	Minutes_Out(0)	5.509	3.976	1.533	10.0	CLK	CLK		0.035
↳ Path 2	-0.871	1	1	4	reg_q(5)_1_7/C	Minutes_Out(5)	5.525	3.964	1.561	10.0	CLK	CLK		0.035
↳ Path 3	-0.862	1	1	1	reg_q(2)_1_10_replica/C	Minutes_Out(2)	5.514	3.981	1.533	10.0	CLK	CLK		0.035
↳ Path 4	-0.862	1	1	6	reg_q(3)_1_9/C	Minutes_Out(3)	5.515	3.959	1.556	10.0	CLK	CLK		0.035
↳ Path 5	-0.854	1	1	1	reg_q(4)_1_8_replica/C	Minutes_Out(4)	5.492	3.959	1.533	10.0	CLK	CLK		0.035
↳ Path 6	-0.848	1	1	1	reg_q(1)_1_11_replica/C	Minutes_Out(1)	5.501	3.968	1.533	10.0	CLK	CLK		0.035
↳ Path 7	-0.748	1	1	7	reg_q(0)_1_18/C	Hours_Out(0)	5.573	3.993	1.580	10.0	CLK	CLK		0.035
↳ Path 8	-0.720	1	1	9	reg_q(1)_1_4/C	Seconds_Out(1)	5.540	3.972	1.568	10.0	CLK	CLK		0.035
↳ Path 9	-0.709	1	1	5	reg_q(4)_1_1/C	Seconds_Out(4)	5.527	3.961	1.566	10.0	CLK	CLK		0.035
↳ Path 10	-0.705	1	1	3	reg_q(4)_1_14/C	Hours_Out(4)	5.529	3.987	1.542	10.0	CLK	CLK		0.035

5. Screenshots of RTL schematics and technology schematics from synthesis

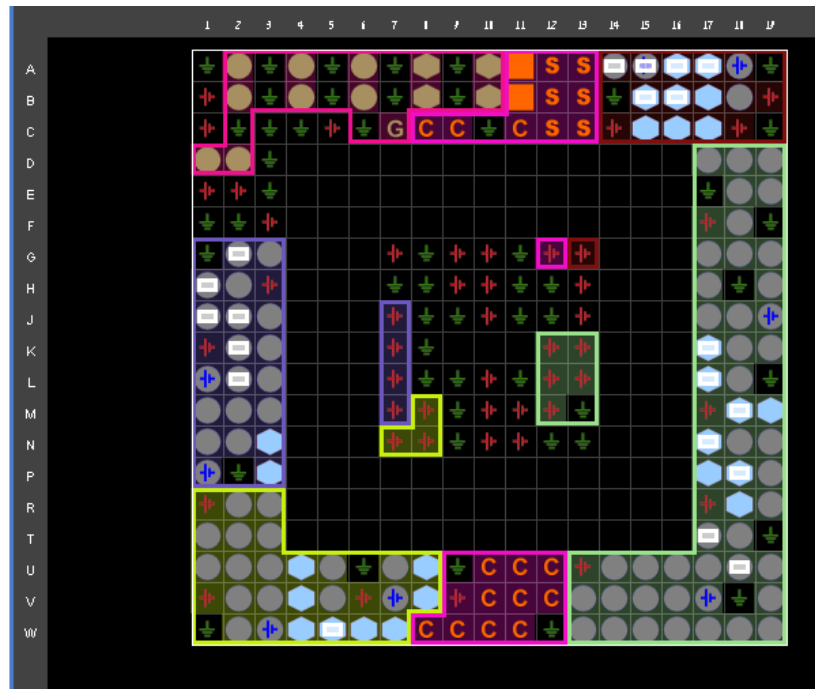
RTL:



technology schematics :



6. Screenshots of device layout and I/O report from Vivado



Name	Direction	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco
All ports (20)							
Hours_Out (5)	OUT			☒	14	LVC MOS33*	3.300
Hours_Out[4]	OUT		L17	☒	14	LVC MOS33*	3.300
Hours_Out[3]	OUT		P18	☒	14	LVC MOS33*	3.300
Hours_Out[2]	OUT		N17	☒	14	LVC MOS33*	3.300
Hours_Out[1]	OUT		M18	☒	14	LVC MOS33*	3.300
Hours_Out[0]	OUT		K17	☒	14	LVC MOS33*	3.300
Minutes_Out (6)	OUT			☒	16	LVC MOS33*	3.300
Minutes_Out[5]	OUT		A17	☒	16	LVC MOS33*	3.300
Minutes_Out[4]	OUT		A15	☒	16	LVC MOS33*	3.300
Minutes_Out[3]	OUT		B16	☒	16	LVC MOS33*	3.300
Minutes_Out[2]	OUT		B15	☒	16	LVC MOS33*	3.300
Minutes_Out[1]	OUT		A16	☒	16	LVC MOS33*	3.300
Minutes_Out[0]	OUT		A14	☒	16	LVC MOS33*	3.300
Seconds_Out (6)	OUT			☒	35	LVC MOS33*	3.300
Seconds_Out[5]	OUT		K2	☒	35	LVC MOS33*	3.300
Seconds_Out[4]	OUT		H1	☒	35	LVC MOS33*	3.300
Seconds_Out[3]	OUT		G2	☒	35	LVC MOS33*	3.300
Seconds_Out[2]	OUT		J2	☒	35	LVC MOS33*	3.300
Seconds_Out[1]	OUT		L2	☒	35	LVC MOS33*	3.300
Seconds_Out[0]	OUT		J1	☒	35	LVC MOS33*	3.300
Scalar ports (3)							
CLK	IN		W5	☒	34	LVC MOS33*	3.300
RESET	IN		U18	☒	14	LVC MOS33*	3.300
TestMode	IN		T17	☒	14	LVC MOS33*	3.300

7. Answers to explicit questions, inspections and measurements required in section E:

ענינו למעלה.